RASHTRIYA BAL VAIGYANIK PRADARSHANI 2025-26

NVS National Level Exhibition

"URBANFLOW : Smart Transportation and Traffic Monitoring System"

Main Theme: Science and Technology for Society

Sub Theme: Communication and Transport

Submitted by,

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SCIENCE EXHIBIT

URBAN FLOW
(SMART TRANSPORT AND TRAFFIC MONITORING SYSTEM)

CERTIFICATE

This is to certify that the project titled **"SMART TRANSPORT AND TRAFFIC MONITORING SYSTEM"** under the Science category has been done by Master**JAYAKRISHNAN S** of class **XII, PM SHRI JAWAHAR NAVODAYA VIDYALAYA, KOLLAM** under my guidance and supervision forcompeting in the national level Science exhibition 2025-26

Guide	
SATHEESH B KUMAR	**PREETHA J**
PGT PHYSICS	**PRINCIPAL**
JNV KOLLAM	**JNV KOLLAM**

ACKNOWLEDGEMENT

I, JAYAKRISHNAN S of Class XII, Jawahar Navodaya Vidyalaya, Kollam, wholeheartedly acknowledge and extend my deepest gratitude to the Principal and teachers of the Vidyalaya for giving me the wonderful opportunity to participate in the national level Science Exhibition 2025-26. This platform has allowed me to showcase my passion for science and technology. I would like to express my profound appreciation and thanks to Mr. Satheesh B Kumar (PGT Physics) for his invaluable guidance and unwavering support throughout the process of creating the exhibit. His expertise and encouragement played a pivotal role in shaping the project and making it a success. I am also grateful to all the staff members and friends who extended their cooperation and assistance in various aspects of the exhibit. His encouragement and constructive feedback motivated me to push my boundaries and achieve the best possible outcome.

Lastly, I want to thank my family for their constant support and belief in my abilities. Their encouragement has been my driving force in pursuing my interests and academic endeavors.

I am truly blessed to have such a supportive and inspiring community around me, and I am determined to continue learning and growing in the field of computer science. Thank you all once again for being an integral part of my journey

Sincerely,

JAYAKRISHNAN S

XII SCIENCE

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INTRODUCTION

In today's rapidly urbanizing world, ensuring smooth mobility and public safety requires intelligent transportation systems and smart infrastructure. UrbanFlow is an innovative IoT-based project developed to address modern urban traffic challenges through automation and real-time responsiveness. Built using the arduino uno and an array of sensors and communication modules, the system integrates multiple functionalities: smart traffic signal control based on vehicle density, emergency vehicle lane prioritization, overloaded vehicle detection, air quality monitoring, and energy-efficient streetlight dimming.

By automating these crucial aspects of traffic management, UrbanFlow reduces congestion, minimizes delays for emergency services, prevents road damage from overloaded vehicles, and ensures eco-friendly operations. Its real-time data-driven decision-making enhances both urban mobility and safety, offering a sustainable and scalable solution for future smart cities. UrbanFlow is not just a model—it's a practical vision for transforming today's traffic systems into intelligent, responsive networks ready for tomorrow.

KEY FEATURES OF THE SYSTEM

1. **Traffic Density-Based Signal Control**
   - The system utilizes ultrasonic sensors to measure the number of vehicles waiting at an intersection. Based on the vehicle count, the traffic signal duration is dynamically adjusted to optimize flow. This reduces unnecessary waiting time and helps decongest busy routes.

2. **Emergency Vehicle Priority Lane**
   - RF transmitter installed in emergency vehicles sends a signal when approaching an intersection.
   - The system detects the signal via an RF receiver and activates a dedicated emergency lane.
   - Simultaneously, halting normal traffic at the intersection , and the LCD displays "EMERGENCY LANE OPEN – RED LIGHT" for safety.

3. **Overloaded Vehicle Detection at Toll**
   - A load cell with HX711 module is used to measure the weight of each vehicle at entry points.
   - If the vehicle exceeds the safe weight threshold, it is flagged as overloaded.
   - The measured weight is shown on the Processing IDE GUI to alert toll authorities in real time.

4. Smart Streetlight Dimming Using Ultrasonic Sensor

- LDR monitors ambient light and activates streetlight control during nighttime or low light conditions.
- Ultrasonic sensors detect approaching vehicles, prompting nearby lights to brighten automatically
- After the vehicle passes, the lights dim again, saving energy during periods of low traffic.

5. Real-Time Data Display with LCD

- A 16x2 I2C LCD screen provides real-time feedback from various modules.
- Displays include traffic signal status, emergency alerts, vehicle weight, and air quality.
- This live data display enhances user interaction and system transparency.

COMPONENTS USED

- **Arduino Uno Microcontroller**
  Acts as the central processing unit of the system. It reads data from sensors and controls output devices such as LEDs, buzzers, relays, and the LCD display based on programmed logic.

- **Breadboard**
  A reusable platform used for building the circuit without soldering. It allows flexible prototyping and easy placement of components and connections.

- **RF Module (433 MHz)**
  Used for wireless communication between modules—for example, transmitting emergency vehicle signals to the traffic controller unit for priority access.

- **Load Cell**
  Detects the weight of vehicles. It helps identify overloaded vehicles, ensuring road safety and preventing infrastructure damage.

- **HX711 Weight Amplifier Module**
  Works with the load cell by amplifying and converting the tiny analog signals into readable digital data for the Arduino to process.

- **Light Dependent Resistor (LDR)**
  Used for controlling streetlights based on ambient light levels. Automatically dims or turns off lights during daytime to conserve energy.

- **Ultrasonic Sensor**
  Measures the distance of approaching vehicles. It is used for traffic density detection and helps in controlling the duration of traffic signals.
- **MQ-135 Air Quality Sensor**
  Monitors the level of harmful gases like CO_2 and NO_2 in the air. It helps assess environmental pollution near traffic zones.
- **LEDs (Red, Yellow, Green)**
  Simulate traffic signal lights, controlled by the Arduino based on traffic density or emergency vehicle presence.
- **LCD Module (16x2)**
  Displays real-time information such as air quality, traffic status, emergency alerts, or overload warnings for easy monitoring.
- **Resistors**
  Regulate current flow in the circuit and protect components like LEDs and sensors from damage due to excess current.

SCIENTIFIC PRINCIPLE INVOLVED

1. Radio Frequency Communication

- **Emergency Vehicle Priority System**: The RF module operates on radio wave transmission, where a transmitter emits a coded signal detected by a receiver. This principle is based on electromagnetic wave propagation and is widely used in wireless communication, remote control, and sensor networks. In this project, it helps identify and prioritize emergency vehicles to ensure safe, uninterrupted passage.

2. Ultrasonic Sound Waves

- **Traffic Density Detection & Smart Streetlight Dimming:** Ultrasonic sensors emit high-frequency sound waves, which reflect off objects and return to the sensor. By measuring the time taken for the echoes to return, the sensor calculates the distance to the object. This principle, known as echolocation, is used in various applications, including medical imaging (ultrasound) and navigator systems.

3. Photoresistance and Light Intensity Detection

- **Smart Streetlight Control:** The Light Dependent Resistor (LDR) operates based on photoresistance — its resistance decreases with increasing light intensity. This principle is applied to detect ambient light levels (day/night) and control streetlight operation. It is widely used in automatic lighting systems and solar tracking applications.

4. Environmental Sensing and Gas Detection

- **Air Quality Monitoring:** The MQ-135 sensor detects gases like CO_2, NH_3, and NOx by measuring changes in conductivity caused by gas adsorption on the sensor's surface. This principle of gas-sensitive semiconductors is fundamental in pollution control, indoor air monitoring, and public health systems.

5. Embedded System and Automation

- **Arduino-Based System Integration:** The project uses microcontroller-based automation, where the Arduino Uno processes sensor inputs and executes programmed control logic. This principle is central to embedded systems engineering, enabling real-time decision-making and actuation in everything from home automation to industrial robotics.

6. Strain Gauge and Wheatstone Bridge Principle

- **Vehicle Overload Detection:** The load cell uses strain gauges that change resistance when weight is applied. These small changes are measured using a Wheatstone bridge circuit. The HX711 module amplifies the signal and sends accurate weight data to the Arduino for processing.

CONSTRUCTION AND WORKING

UrbanFlow is built using five Arduino Uno microcontrollers to independently manage each core module, ensuring smooth, parallel operation across the **traffic control**, **emergency response**, **load detection**, **streetlight dimming**, and **air quality monitoring systems**. Each module was tested individually and then integrated into a central system to ensure modularity and real-time responsiveness.

Here's a basic outline of how the **UrbanFlow**works with arduino, and other sensors :

1. **Traffic Density-Based Signal Control with Emergency Vehicle Priority**
   - **Components**: 3 Ultrasonic Sensors, RF Receiver, LEDs (Red, Yellow, Green), Buzzer, LCD Display, Arduino Uno
   - **Setup**:Ultrasonic sensors are installed at three lanes of an intersection to count the number of waiting vehicles. An RF receiver is placed near the intersection to detect emergency vehicle signals. LEDs simulate traffic lights, and an LCD displays real-time signal status.
   - **Principle**: The ultrasonic sensors detect vehicle presence based on distance. The Arduino calculates traffic density and allocates green light durations dynamically to reduce congestion. When an emergency vehicle transmits a signal via RF, the system overrides normal timing logic. It halts regular traffic, activates a dedicated emergency lane (green LED), triggers the buzzer, and displays "EMERGENCY LANE OPEN – RED LIGHT" on the LCD

for safety and awareness.

2. **Overloaded Vehicle Detection**

   - **Components**: Load Cell, HX711 Module, Arduino Uno
   - **Setup**: The load cell is embedded in a platform where vehicles are weighed before toll or bridge entry. The HX711 module amplifies the signal and sends data to the Arduino, which transmits it to a computer via serial connection for display on Processing IDE.
   - **Principle**: When a vehicle enters the platform, its weight is measured by the load cell. If the reading exceeds the predefined limit, the vehicle is flagged as overloaded. The real-time weight data and alert are displayed on the GUI, aiding toll officers in making informed decisions.

3. **Smart Streetlight Dimming System**

   - **Components**: LDR, Ultrasonic Sensor, LEDs (Streetlights), Arduino Uno
   - **Setup**: An LDR detects ambient lighting conditions, and an ultrasonic sensor is positioned roadside to sense approaching vehicles. LEDs represent streetlights.
   - **Principle**: During dark hours (low LDR reading), the streetlight system is activated. When a vehicle approaches, the ultrasonic sensor detects movement and the corresponding LEDs brighten. After the vehicle passes, the lights return to dim mode, reducing power usage while maintaining visibility.

4. Air Quality Monitoring System

- **Components**: MQ-135 Air Quality Sensor, LCD Display, Arduino Uno
- **Setup**: The MQ-135 sensor is mounted near the simulated road to monitor air quality. Data from the sensor is processed by the Arduino and shown on an LCD screen.
- **Principle**: The sensor detects gas concentrations (CO_2, NH_3, NOx) and outputs a signal based on the pollutant level. The Arduino converts this into readable values and classifies the air quality as **Fresh**, **Moderate**, or **Polluted**, helping raise awareness of environmental conditions in traffic zones.

WORKING MECHANISM

1. **Traffic Density-Based Signal Control with Emergency Vehicle Priority**

    - Ultrasonic sensors continuously detect the number of vehicles in each lane by measuring the distance to the closest object.
    - Based on the vehicle count, the Arduino adjusts the green light timing to prioritize high-traffic lanes, reducing congestion.
    - When an emergency vehicle transmits a signal using RF, the RF receiver activates emergency mode: all lanes are stopped, a dedicated lane gets green light, a buzzer alerts nearby vehicles, and the LCD shows a warning.

2. **Overloaded Vehicle Detection**

    - The load cell senses the vehicle's weight when it moves onto the weighing platform.
    - The signal from the load cell is amplified by the HX711 module and processed by the Arduino.
    - If the weight exceeds the set limit, the system flags it as overloaded and displays the reading and alert on the GUI (Processing IDE).

3. Smart Streetlight Dimming System

- The LDR detects ambient light levels; during low light (evening or night), the system is activated.
- When a vehicle approaches, the ultrasonic sensor detects motion and instructs the LEDs (streetlights) to brighten.
- Once the vehicle passes, the lights automatically dim, ensuring energy-efficient lighting without compromising visibility.

4. Air Quality Monitoring System

- The MQ-135 sensor monitors the concentration of harmful gases like CO_2, NH_3, and NOx.
- The Arduino processes this data and classifies the air quality level as **Fresh**, **Moderate**, or **Polluted**
- The current air quality status is continuously displayed on an LCD, promoting environmental awareness in urban areas.

UrbanFlow is an advanced, IoT-based traffic management system that integrates multiple modules into a cohesive platform aimed at improving urban mobility, road safety, and sustainability. The system combines traffic density-based signal control, emergency vehicle priority access, overloaded vehicle detection, smart streetlight dimming, and air quality monitoring. Each of these modules works independently but is orchestrated together through centralized microcontroller units, primarily Arduino Uno boards, ensuring real-time responsiveness and accurate decision-making across all traffic scenarios.
Always follow the safety guidelines while working withsensors and devices.

The centralized Arduino units process sensor inputs, control relays, manage signals, and display alerts on LCDs or buzzers. This coordination enables smooth traffic flow, emergency access, environmental monitoring, and efficient streetlight usage—all working together for better city management.

In summary, **UrbanFlow** is more than just a student project—it's a blueprint for smart city traffic systems. By automating core infrastructure functions and enabling real-time interaction between vehicles, roads, and environmental elements, the system minimizes human intervention, maximizes resource efficiency, and contributes to a cleaner, safer, and smarter urban future.

ADVANTAGES OF MY PROJECT

1. **Improved Traffic Flow and Emergency Access**
   - The integration of traffic density detection and emergency vehicle priority ensures optimized traffic signal timing. Emergency lanes automatically clear paths for ambulances and fire trucks, reducing response time and preventing congestion.
2. **Energy Efficiency through Smart Lighting**
   - Smart streetlight dimming conserves electricity by adjusting brightness based on vehicle presence using sensors. This reduces energy waste during low-traffic hours and promotes sustainable urban infrastructure.
3. **Enhanced Urban Safety and Monitoring**
   - The overloaded vehicle detection module helps identify rule violations and protects road infrastructure. Additionally, continuous air quality monitoring ensures environmental awareness, enabling cities to take prompt action against pollution.
4. **Real-Time Automation and Control**
   - With Arduino Uno at its core, UrbanFlow ensures real-time data processing, signal control, and feedback through LEDs, LCDs, and buzzers—minimizing manual work while increasing system responsiveness and efficiency.

5. Modular and Scalable Design

- UrbanFlow is built using independent modules that can be easily modified or scaled. This modular approach allows the system to be upgraded with new technologies or expanded to manage larger city areas without redesigning the entire setup.

6. Supports Smart City Development

- By integrating automation, sensor-based decision-making, and real-time monitoring, UrbanFlow lays the groundwork for future smart city initiatives. It enhances public service delivery, reduces human errors, and promotes intelligent urban infrastructure

DISADVANTAGES OF MY PROJECT

- **Weather Sensitivity**: Components like LDRs and ultrasonic sensors can be affected by rain, fog, or excessive sunlight, potentially disrupting system performance in outdoor environments.

- **Sensor Limitations**: Ultrasonic and IR sensors have a restricted detection range, which may reduce accuracy in large or congested urban areas, leading to occasional false readings or missed vehicles.

CONCLUSION

In conclusion, **UrbanFlow** is a comprehensive and innovative solution for modern urban transport challenges. It leverages sensor-based automation and Arduino-controlled modules to create a responsive, real-time traffic management system. By combining multiple functionalities into a single system, UrbanFlow significantly enhances road safety, traffic efficiency, and emergency response.

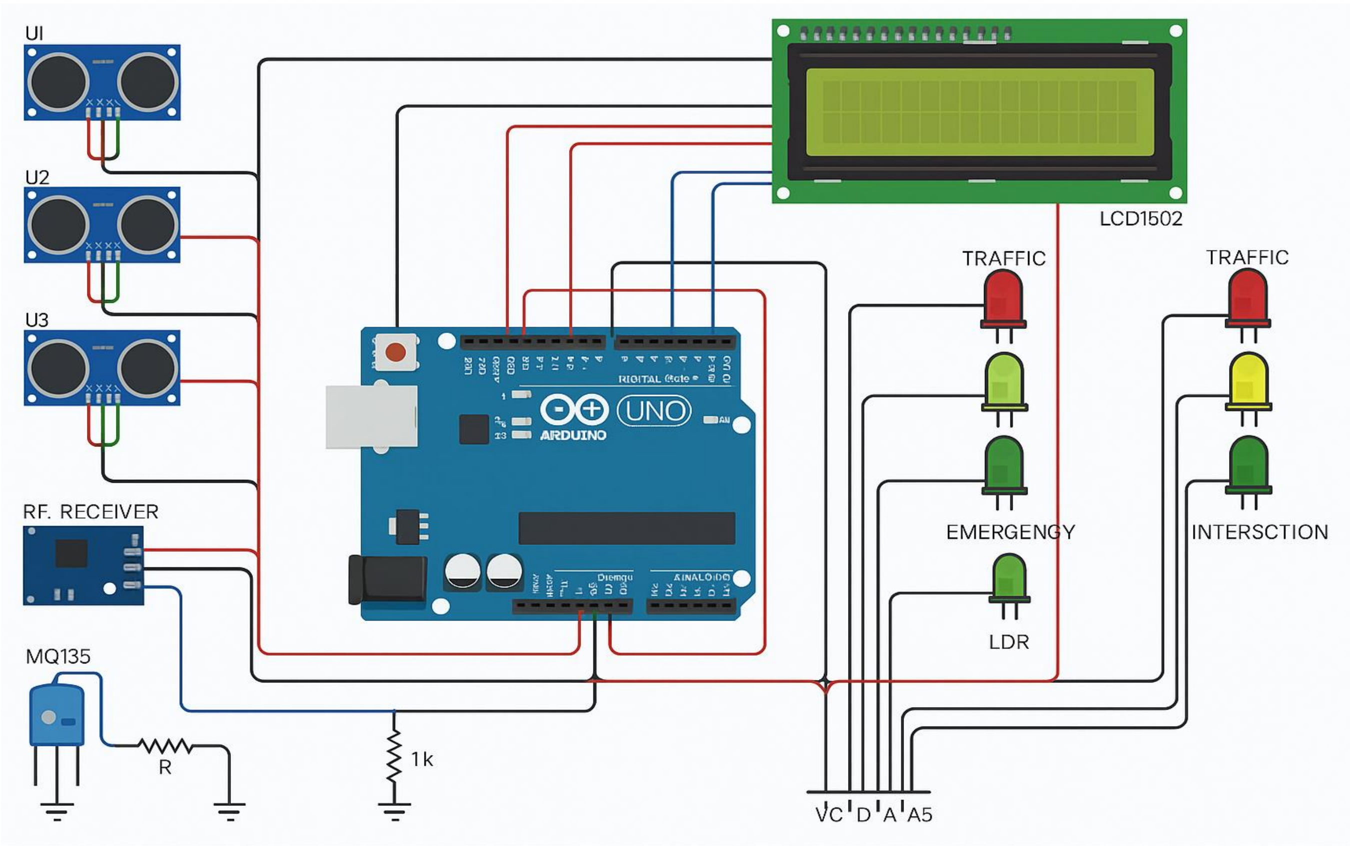
The system’s standout feature is its **integrated traffic signal and emergency vehicle management**, which uses ultrasonic sensors to detect traffic density and automatically adjust signal timings. When an emergency vehicle is detected, it prioritizes access, ensuring faster response times without manual intervention. This smart mechanism reduces congestion and keeps critical services moving swiftly.

UrbanFlow also addresses safety and regulation through **overload detection**, identifying vehicles exceeding weight limits using ultrasonic sensors. Simultaneously, the **smart streetlight dimming module** helps optimize energy usage by adjusting brightness based on ambient lighting, contributing to cost-effective city operations.

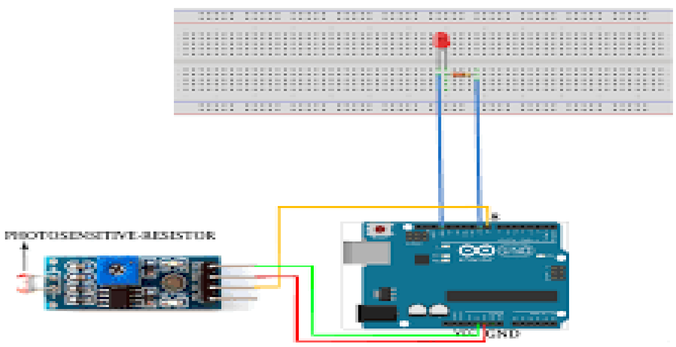
Altogether, UrbanFlow exemplifies how low-cost automation can be effectively applied in urban infrastructure. Its modular design, real-time responsiveness, and minimal maintenance requirements make it a scalable solution for future **smart city traffic systems**.

ILLUSTRATIONS

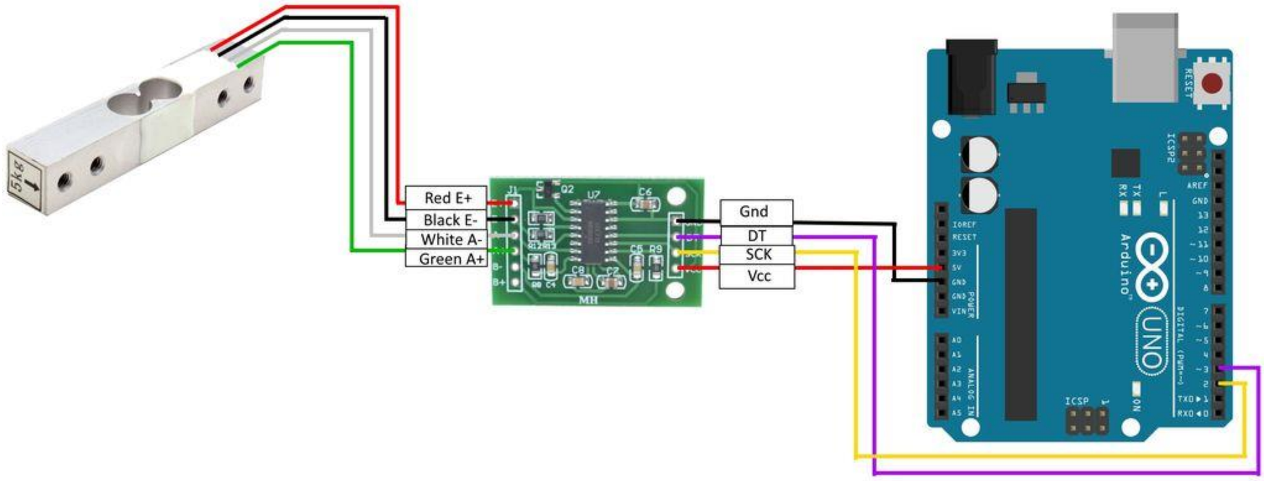
1.TRAFFIC DENSITY AND EMERGENCY DETECTION WITH ARDUINO



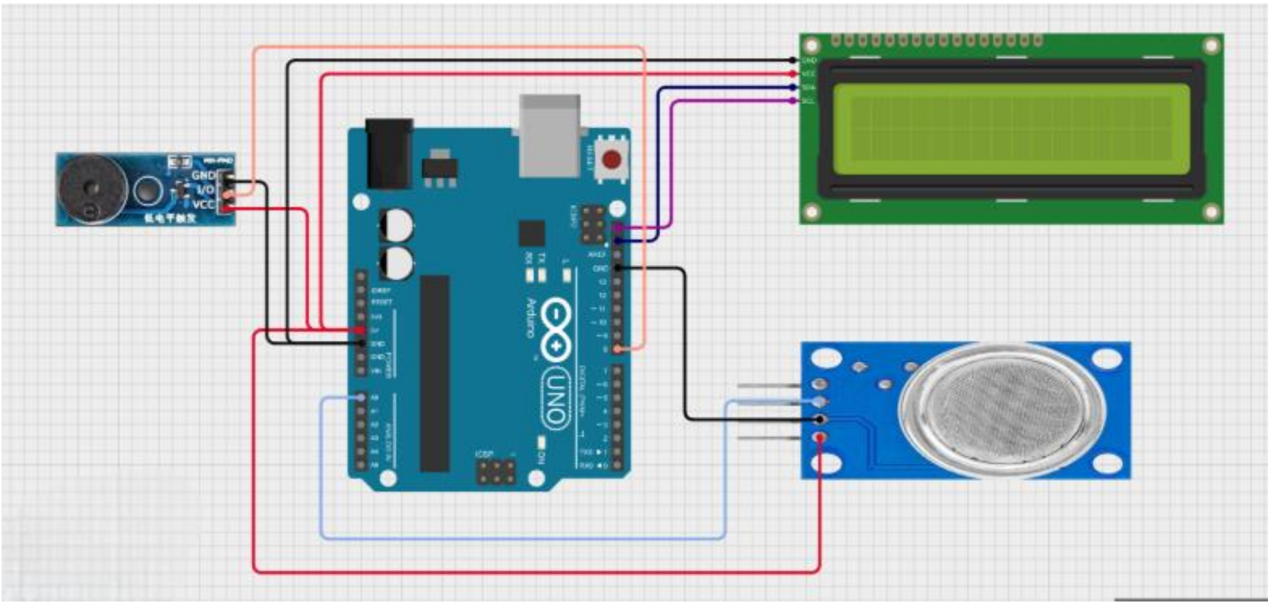
2. LIGHT AUTOMATION DAYNIGHT WITH DIMMING



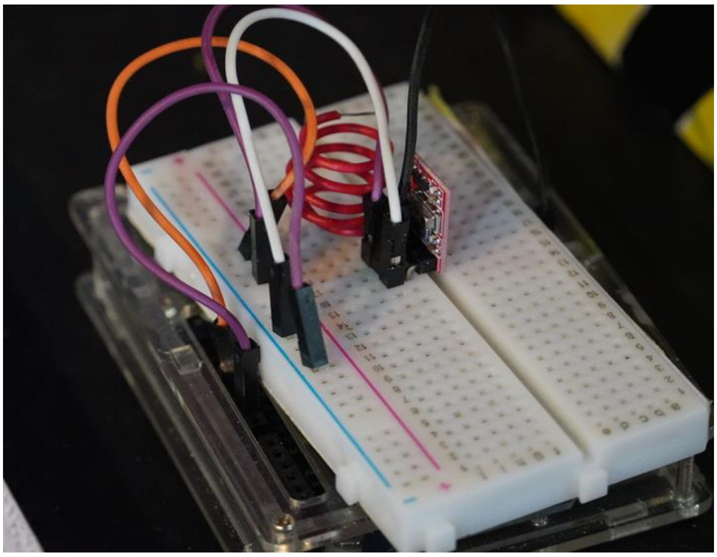
3.OVERLOADED VEHICLE DETECTION

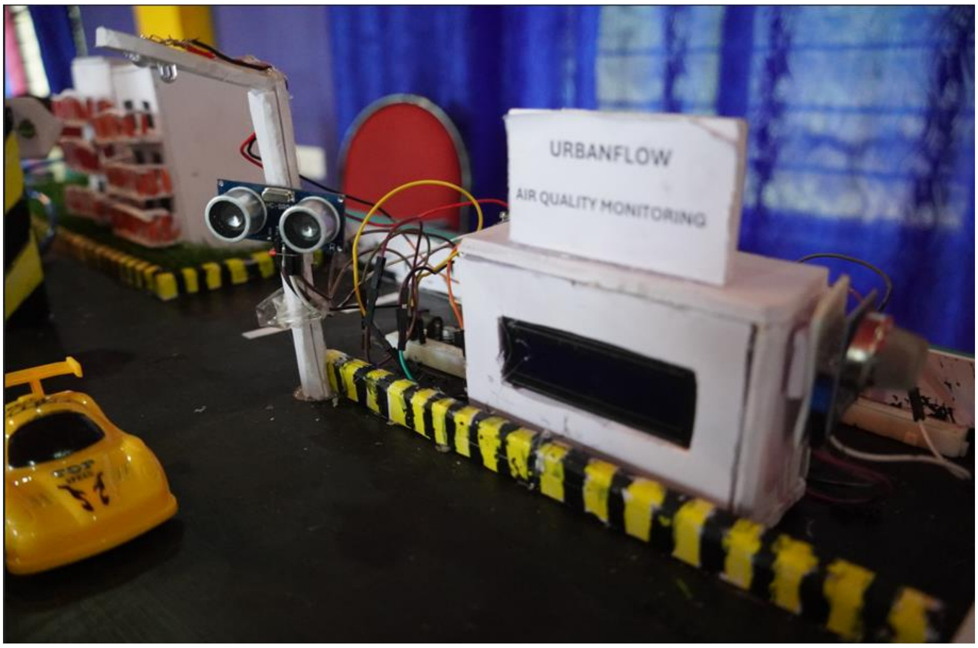


4.AIR QUALITY MONITORING



PHOTOGRAPH OF THE EXHIBIT




URBANFLOW
AIR QUALITY MONITORING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